

Project Title: Fitness Market Pricing Analytics: A Cross-Border
Comparative Study

Author: Rimas Almontashiri

Role: Data Scientist / Business Intelligence Analyst

Location: Jeddah, Saudi Arabia

Professional Profiles: www.linkedin.com/in/rmsmunti

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1. Introduction & Background

In the field of data science, real-world problems provide the most compelling foundations for market analysis. This independent research project was initiated to investigate the structural pricing methodologies of modern fitness club networks. For demographic segments with fixed budgets, such as university students, subscription fees represent a significant financial variable.

This report delivers an end-to-end data analysis evaluating the subscription matrices of dominant market players across three distinct economic territories: Saudi Arabia (KSA), the United Arab Emirates (UAE), and the United Kingdom (UK). The objective is to decode how corporate entities leverage localized demographics, tier architectures, and service bundling to drive revenue optimization.

A. Rationale for Project Scope (PureGym & GymNation)

Two dominant fitness chains were meticulously selected to establish a fair and logical market benchmark:

1. **PureGym:** Selected because it is a massive global brand operating directly in both the UK market and the local Saudi market. This provides a direct window to track how the exact same parent brand alters its pricing behavior between two completely different macroeconomic environments.
2. **GymNation:** Selected because it represents the most prominent regional competitor in the UAE, utilizing an identical corporate philosophy of "high-value, low-cost" fitness. Comparing GymNation with PureGym provides an ideal competitive baseline to analyze pricing dynamics within closely linked regional and international landscapes.

2. Technical Methodology & Data Lifecycle

A. Data Collection Process (Web Scraping & Extraction)

To build an authentic, rare non-traditional dataset, a customized data aggregation process was designed to crawl and extract membership variables directly from corporate environments:

- **Data Sources:** Live public subscription data extracted from the official portals of PureGym KSA, PureGym UK, and GymNation UAE.
- **Collection Strategy:** Utilized specialized data extraction methodologies to programmatically isolate current subscription tiers, monthly fees, registration costs, and itemized feature matrices.

B. Data Cleaning & Statistical Filtering Criteria

To guarantee a strict "apple-to-apple" comparison and prevent mathematical distortion across datasets, rigorous filtering and preprocessing were executed manually:

1. **Exclusion of Off-Peak Memberships:** Off-peak subscriptions were entirely removed from the scope. Because these plans restrict gym access to low-traffic, limited hours of the day, including them would artificially deflate statistical averages when compared against standard 24/7 access plans.
2. **Exclusion of Long-Term Commitments:** Subscriptions requiring annual or long-term contract lock-ins were excluded. Long-term contracts alter baseline monthly fees in exchange for penalty clauses; therefore, the analysis isolates flexible, month-to-month, non-commitment contracts available to all consumers.
3. **Manual Currency Normalization:** To maintain absolute data integrity and tight control over pre-processing, all foreign currency metrics—British Pounds (GBP) and UAE Dirhams (AED)—were manually converted into Saudi Riyals (SAR) using real-time baseline conversion ratios before ingestion, ensuring a fully unified analytical baseline.

C. Automated Global Price Gap Analytics (Python Framework)

The analytical core of the Python workflow focused on building a robust, fault-tolerant evaluation script to mathematically map pricing variances across global markets.

Instead of relying on rigid, prone-to-error tabular groupings, a dynamic analytical function was developed to handle potential string inconsistencies in the extracted text. The algorithm programmatically enforced lower-case standardization and stripped hidden whitespace padding across the 'Brand' and 'Tier' dimensions. This step ensured that cross-border calculations targeted exact matching entities securely.

The framework dynamically isolated subset populations for targeted markets, computed mean distributions safely, handled potential mathematical execution errors, and calculated both relative percentage gaps and absolute monthly deviations (SAR/mo) across comparable membership tiers.

Algorithmic Outputs & Empirical Findings:

1. PureGym Benchmarks (KSA vs. United Kingdom):

- **Core Tier:** The automated analytics isolated an extreme geographic pricing delta, proving that KSA is 58.9% MORE EXPENSIVE than the UK, representing an absolute markup of +73.8 SAR/month.
- **Plus Tier:** A parallel expansion was captured in the premium bracket, where KSA is 55.3% MORE EXPENSIVE than the UK, translating to an extra +88.7 SAR/month.

2. GymNation Benchmarks (KSA vs. UAE - Dubai):

- **Core Tier:** The regional comparison indicated tight integration, with KSA yielding a slight variation of 2.2% CHEAPER than Dubai (-4.8 SAR/month).
- **Plus Tier:** Equilibrium shifts marginally in the mid-tier, showing KSA is 1.8% MORE EXPENSIVE than Dubai (+4.5 SAR/month).
- **Signature Tier:** The elite bracket aligns back downward, with KSA scaling at 2.2% CHEAPER than Dubai (-9.1 SAR/month).

These empirical findings mathematically demonstrate that while regional Gulf markets (KSA and UAE) are highly synchronized and priced symmetrically, global operators enforce aggressive localized markups when deploying operations domestically compared to European baselines.

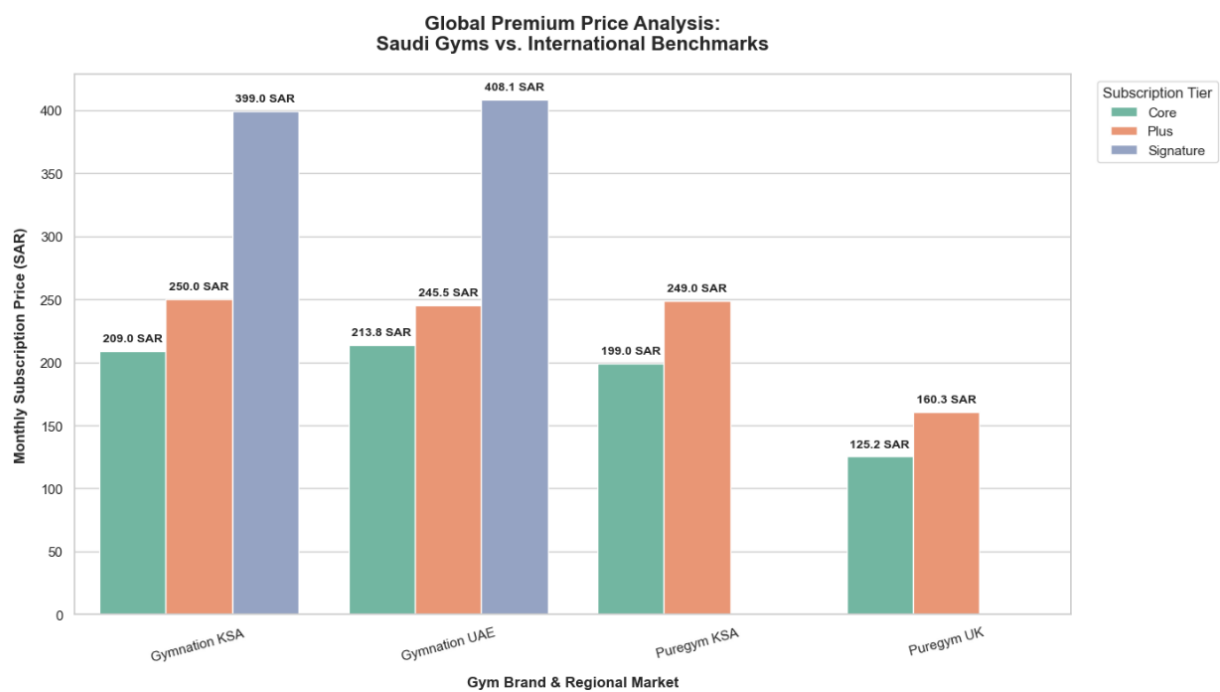


Figure 1: Python Exploratory Data Analysis displaying pricing distributions and variance across extracted entities.

D. Interactive Analytical Modeling (Power BI)

Following the data engineering phase, the clean dataset was modeled into an interactive enterprise dashboard using Power BI:

- **Dynamic DAX Architecture:** Advanced Data Analysis Expressions (DAX) were authored to calculate dynamic KPIs, establishing a precise Global Price Average of 245.89 SAR.
- **Visual Optimization:** Implemented a dark-mode thematic interface incorporating interactive Line and Donut charts alongside a comprehensive Gym Tiers Feature Comparison Matrix to evaluate service delivery against consumer financial input.

E. Methodological Innovation: Embracing "Vibe Coding"

This project intentionally departs from rigid, traditional development cycles by adopting Vibe Coding—a modern paradigm where the data scientist operates as a high-level architect and logic designer, while generative AI models drive the programmatic execution.

- **The Framework:** Instead of spending hours debugging syntax or manual boilerplate formatting, human intellect was directed toward framing the market logic, optimizing the statistical parameters, and engineering the analytical layout.
- **The Execution:** Leveraging advanced LLMs, the data scraping scripts, custom DAX metrics, and automated gap analytics were rapidly iterated and refined. This approach proves how modern data professionals can utilize collaborative AI to deliver end-to-end, enterprise-grade data products with maximum velocity and strategic focus.

3. Core Market Insights & Pricing Strategies

The analytical pipeline successfully uncovered three foundational pricing strategies deployed by the target fitness chains:

A. Geographical Price Discrimination

The empirical evidence indicates that subscription pricing is heavily dictated by localized market positioning and regional purchasing power rather than uniform global operational costs.

- **Analysis:** Evaluating the baseline Core Tier under identical corporate umbrellas reveals a significant geographical disparity. PureGym UK maintains a localized market average of 142.76 SAR, whereas PureGym KSA scales sharply to 224.00 SAR for identical baseline facility access.

B. Tiered Pricing Framework

Gym operators strictly avoid flat-rate monetization, engineering structured tiers (Core, Plus, and Signature) to effectively capture diverse consumer segments.

- **Analysis:** Entry-level tiers minimize user acquisition friction, while premium upgrades extract maximum consumer surplus. This strategy is highlighted by the Signature Tier within the UAE market (GymNation UAE), which peaks at an average of 408.11 SAR by targeting the premium consumer segment.

C. Value-Based Service Bundling

The Matrix Visual isolates how chains systematically justify price appreciation between membership tiers through strategic utility coupling.

- **Analysis:** Price escalation directly correlates with high-perceived-value add-ons. Standard tiers restrict access strictly to the primary facility, whereas top-tier structures bundle services like Multi-Club Access, Guest Privileges, and Body Composition Analysis to organically incentivize users to migrate to premium monthly brackets.

4. Visualizing the Market: Dashboard Deployment

To facilitate real-time insight discovery, the final analytical layers were deployed via an interactive executive dashboard.

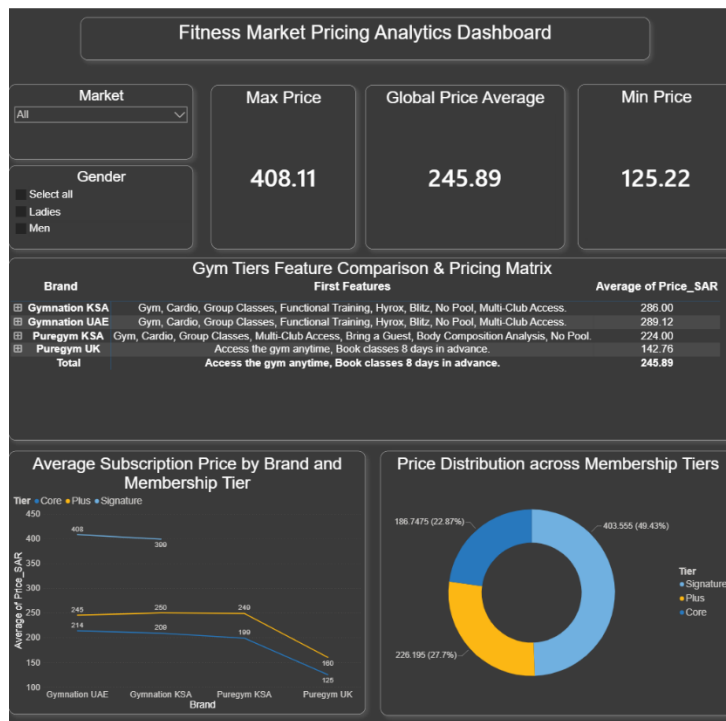


Figure 2: Fitness Market Pricing Analytics Dashboard featuring automated KPIs, tier distributions, and feature matrix.

5. Conclusion & Industry Recommendations

This independent data study demonstrates that international fitness operators scale their pricing architectures dynamic to regional parameters, tier segmentation, and value bundling.

Strategic Recommendations:

- 1. Market Penetration via Structured Tiering:** To capture the highly recurring student demographic without diluting premium brand equity, regional operators should introduce specialized, feature-stripped subscription plans.
- 2. Value-Driven Monetization:** Operators must carefully structure their Matrix Bundles, ensuring that high-margin add-ons (such as digital body metrics) are placed exclusively in premium tiers to justify the price-to-value gap to the consumer.

Future Work:

KSA Market Deep Dive (Phase 2 Extension):

While this initial phase establishes a global pricing benchmark across borders, a subsequent phase is planned to isolate and deeply analyze the domestic Saudi Arabian fitness ecosystem. The upcoming extension will leverage this clean baseline to evaluate internal corporate mechanisms, focusing on:

- 1. Gender-Based Pricing Variance:** Investigating potential pricing disparities between Ladies' and Men's membership segments within KSA.
- 2. Utility vs. Cost Modeling:** Mapping localized service and feature bundling against subscription price appreciation.

3. Temporal Flexibility Analytics: Re-introducing off-peak and commitment models to decode how operators utilize time-bound availability to drive premium monetization.